Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CSDE733/18ISDE732]

SEVENTH SEMESTER B. Tech EXAMINATION, DECEMBER–2021

WEB SEMANTICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	With neat diagram, explain layered approach of semantic web.	**07**
	b.	Explain Semantic web technologies in brief.	**07**
2.	**a.**	Illustrate the fundamental concepts of RDF.	**07**
	b.	Discuss XML based syntax of RDF in detail.	**07**
3.	**a.**	Explain the layering of ontology web Language(OWL).	**07**
	b.	Explain monotonic rules with suitable example.	**07**
		OR	
4.	**a.**	List and explain the possible future extensions of ontology web Language(OWL).	**07**
	b.	Explain the process of encoding in XML.	**07**
5.	**a.**	Illustrate the problems and the contribution of Semantic Web technology with respect to web services.	**07**
	b.	Analyze the various problems that can overcome by Semantic web.	**07**
		OR	
6.	**a.**	Describe Horizontal information products from Elsevier.	**07**
	b.	Explain Thinktank portal at EnerSearch.	**07**
7.	**a.**	List and explain any three main stages for ontology development.	**07**
	b.	Illustrate the various types of ontologies that supported using ML-style techniques.	**07**
		OR	
8.	**a.**	With neat diagram, explain On-To-Knowledge architecture.	**07**
	b.	Describe re-use of existing technologies.	**07**